

C. Documentations

C1. Technical Documentation

1. Project Overview

FirstAid Connect is a Power Apps Canvas App designed for a community first aid organisation. The app connects three user groups:

Role	Main Purpose
Resident / Public	Submit emergency SOS requests and access first aid guidance.
Administrator	Monitor incidents, view volunteer availability, dispatch volunteers, and create shifts.
Volunteer	View current status, check assigned shifts, open assigned incident details, and mark incidents as resolved.

The system was developed and improved from the Assignment 1 prototype into a functional Power Apps prototype. The original prototype already defined role-based navigation for Resident, Volunteer, and Administrator users, including Emergency SOS, First Aid Guide, Dashboard, Incident Detail, Create Shift, and Dispatch pages.

2. System Architecture

The app uses a simple Power Apps + Microsoft Lists architecture.

Layer	Technology	Purpose
Frontend	Microsoft Power Apps Canvas App	Provides mobile-style user interface and role-based workflows.
Data Backend	Microsoft Lists / SharePoint Lists	Stores incidents, volunteers, and shifts.
Logic Layer	Power Fx formulas	Handles navigation, validation, filtering, counting, Patch updates, and status changes.
Testing Tools	Power Apps Test Studio + Monitor	Used for automated testing and runtime performance observation.
Source Control	GitHub	Stores unpacked Power Apps source code and documentation evidence.

3. Data Backend Design

The backend contains three main SharePoint Lists:

- 1

Incidents
- 2

Volunteers
- 3

Shifts

3.1 Incidents List

The `Incidents` list stores emergency reports submitted by residents and updated by administrators/volunteers.

https://unisydneymy.sharepoint.com/:g/personal/sili0424_uni_sydney_edu_au/JADxLkK_UMPfQ46ZVaosuDxcASL-XwVsieH3GBCXJjbgWv0?e=6swlp9 Connect your OneDrive account

lists

idents ☆

All Items

	IncidentID	Types	Location	Priority	Notes	Status	AssignedVolunteer	CreatedTime	+ Add column
INC-20260512-001	INC-20260512-001	Cardiac	Near Main Stage	Critical	Elderly male with chest pain	Resolved	Sarah O'Brien	6 days from now	
INC-20260512-002	INC-20260512-002	Laceration	Food Court Area B	Moderate	Deep cut from broken glass	Resolved	Liam Chen	Tuesday at 9:09 PM	
INC-20260512-003	INC-20260512-003	Heat Stress	Camping Zone 3	Moderate	Patient dizzy and dehydrated	Resolved	James Kim	4 days from now	
INC-20260508-004	INC-20260508-004	Laceration	usyd	Critical	hurt	Resolved		Yesterday at 10:07 ...	
INC-20260508-005	INC-20260508-005	SOS Emergency	unsw	Critical	leg and arm hurt	Resolved	Aisha Patel	Yesterday at 11:05 ...	
INC-20260509-006	INC-20260509-006	SOS Emergency	usyd	Critical	leg and arm hurt	Assigned	Liam Chen	4 minutes ago	
INC-20260509-007	INC-20260509-007	Heat Stress	TownHall	Critical	Patient dizzy	New		A few seconds ago	

3.2 Volunteers List

The **Volunteers** list stores volunteer profiles, status, sector, and active task information.

https://unisydneyedu-my.sharepoint.com/:l:/g/personal/sili0424_uni_sydney_edu_au/JACrm8RN6U-URKkq9erLKHNRAYrYITd74vh7C05VKRXTFgI?e=zhKMb3 [Connect your OneDrive account](#)

lists

lunteers ☆

All Items

unteerID	Name	Role	Sector	Status	CurrentTask	+ Add column
01	Liam Chen	First Aider	Main Stage	Responding	INC-20260509-006	
02	Sarah O'Brien	Paramedic	Food Court	Available		
03	James Kim	First Aider	Entry Gate	Available		
04	Emily Davis	Nurse	Camping Area	On Break		
05	Aisha Patel	Paramedic	Roaming	Available		

3.3 Shifts List

The **Shifts** list stores scheduled volunteer duties created by administrators.

https://unisydneyedu-my.sharepoint.com/:l:/g/personal/sili0424_uni_sydney_edu_au/JACLvoIPHqh7SI8yThqBJ8SWAWuxBAorK6i3uJXMJqjbUbs?e=jMd5ud [Connect your OneDrive account](#)

lists

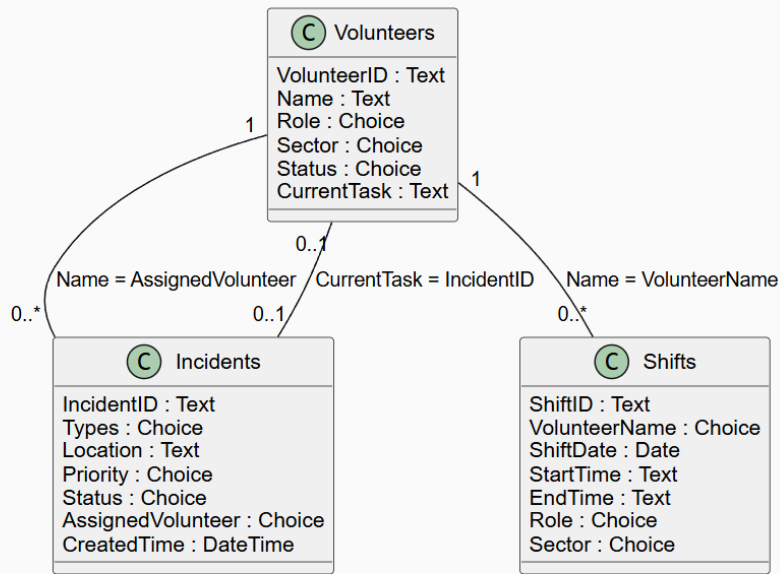
ifts ☆

All Items

	ShiftID	VolunteerName	ShiftDate	StartTime	EndTime	Role	Sector	+ Add column
01	S001	Liam Chen	4 days from now	08:00	14:00	First Aider	Main Stage	
02	S002	James Kim	6 days from now	14:00	20:00	First Aider	Entry Gate	
03	S003	Aisha Patel	May 16	08:00	14:00	Paramedic	Roaming	
INC-20260508-004	S-20260508-004	Sarah O'Brien	Yesterday at 7:00 AM	09:00	11:00	First Aider	Main Stage	
INC-20260508-005	S-20260508-005	Sarah O'Brien	Yesterday at 7:00 AM	12:00	18:00	First Aider	Food Court	
INC-20260509-006	S-20260509-006	James Kim	13 hours ago	08:00	14:00	Team Lead	Roaming	

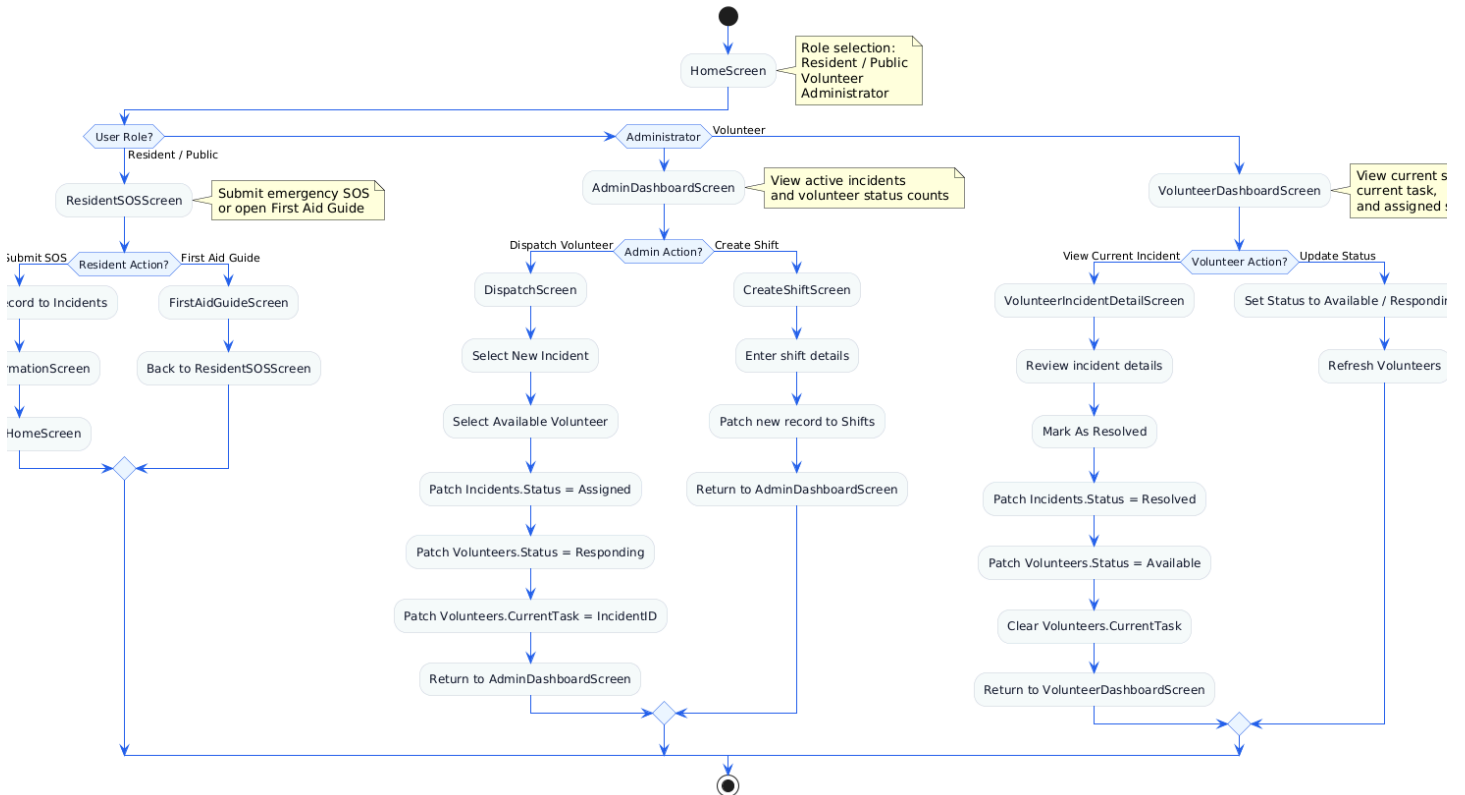
4. Data Backend Diagram

FirstAid Connect - Data Backend Diagram



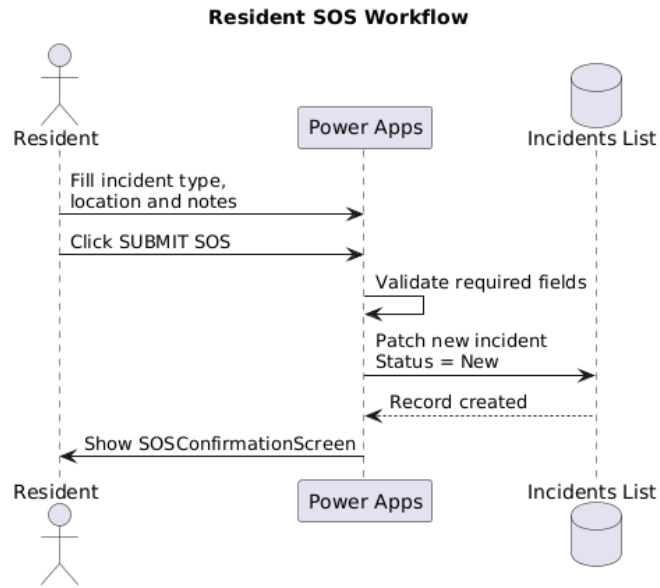
5. UI Flow Diagram

FirstAid Connect - UI Flow Diagram

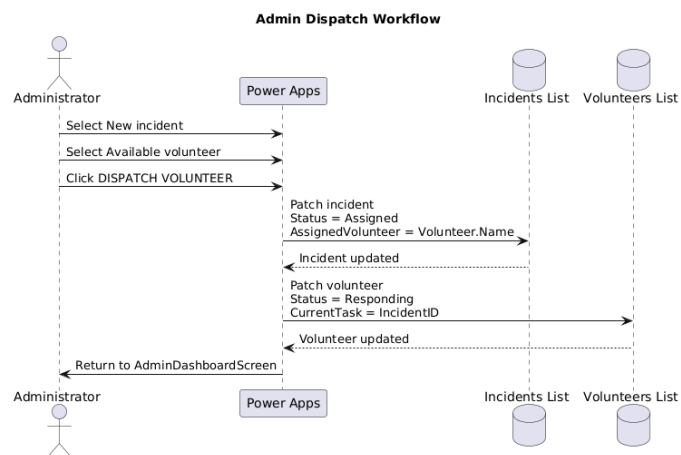


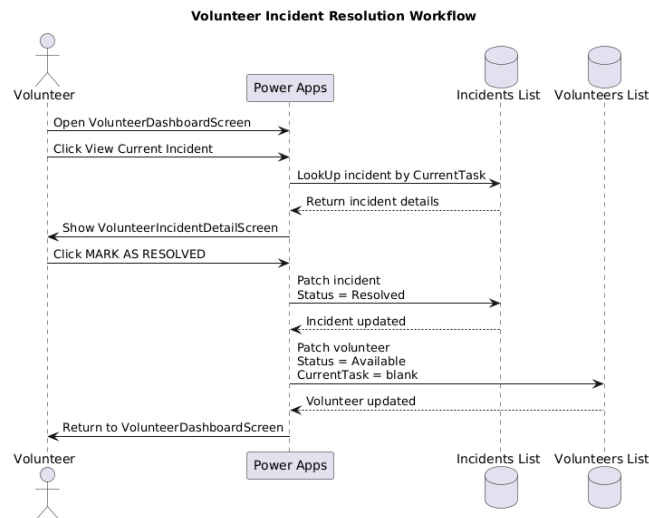
6. Main Data Workflows

6.1 Resident SOS Workflow



6.2 Dispatch Workflow





C2 FirstAid Connect Volunteer How-to Guide

Purpose of this Guide

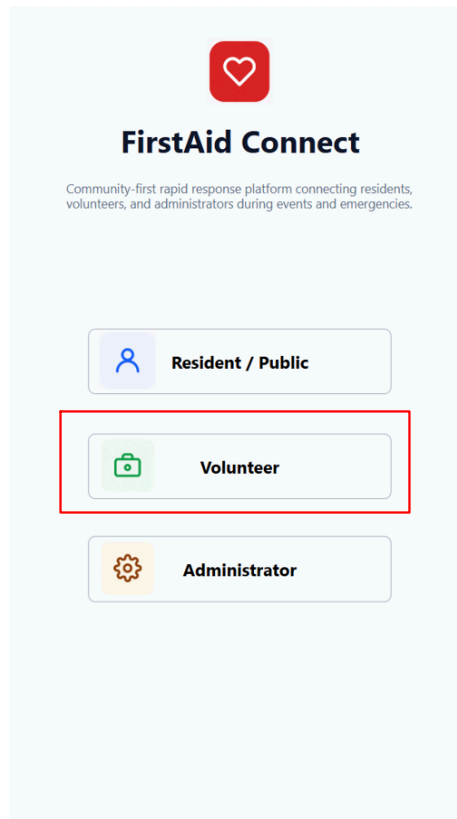
This guide is designed for community safety volunteers using **FirstAid Connect** in the field. It explains how to check your volunteer status, view assigned incidents, review emergency details, and update the system after responding.

FirstAid Connect supports fast coordination between residents, administrators, and volunteers during community events and emergency situations.

1. Open the App

When you open FirstAid Connect, you will see the main role selection screen.

Figure 1. FirstAid Connect Home Screen



On this screen, choose your role:

Button	Use
Resident / Public	Used by members of the public to submit SOS requests.
Volunteer	Used by field volunteers to check shifts and incident assignments.
Administrator	Used by coordinators to monitor incidents and dispatch volunteers.

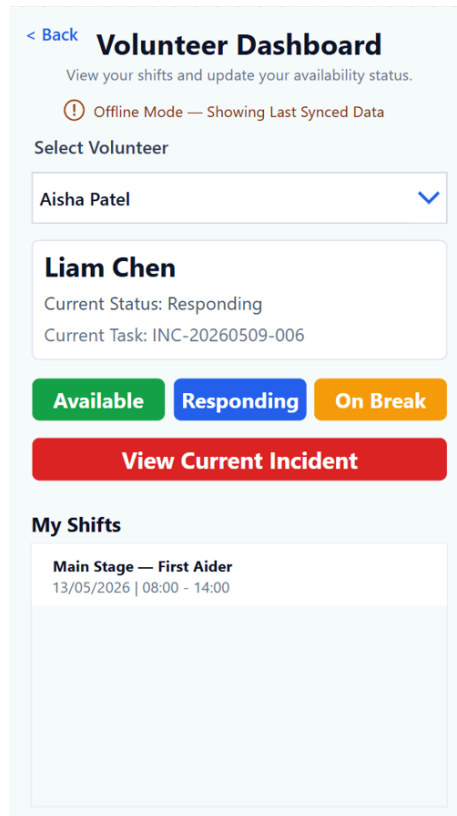
As a community safety volunteer, tap:

1 Volunteer

2. Open the Volunteer Dashboard

After selecting **Volunteer**, you will arrive at the **Volunteer Dashboard**.

Figure 2. Volunteer Dashboard



The Volunteer Dashboard allows you to:

- 1 - Select your volunteer profile
- 2 - Check your current availability status
- 3 - View your current assigned incident
- 4 - Update your status
- 5 - View your upcoming shifts

4. Understand Your Volunteer Status

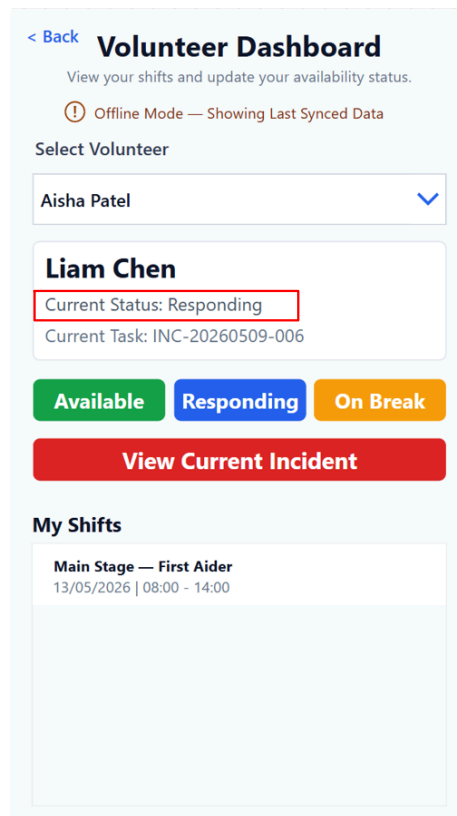
Your current status is displayed in the profile card.

Status	Meaning
Available	You are ready to be assigned to an incident.
Responding	You are currently responding to an assigned incident.
On Break	You are temporarily unavailable.

The three status buttons allow you to update your availability:

- 1 Available
- 2 Responding
- 3 On Break

Use these buttons carefully, because administrators use your status to decide who can be dispatched.



5. Update Your Availability

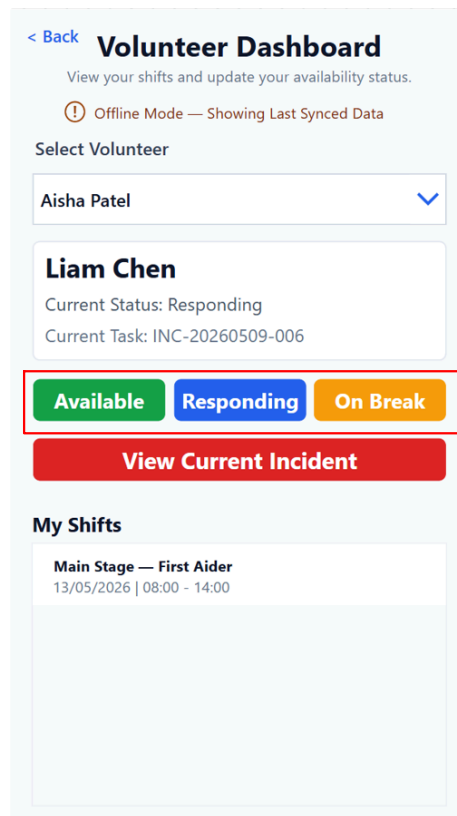
Figure 4. Volunteer Status Buttons

Insert screenshot showing Available / Responding / On Break buttons

![VolunteerStatusButtons]

To update your status:

1. Open the Volunteer Dashboard.
2. Select your volunteer profile.
3. Tap one of the status buttons:
 - **Available**
 - **Responding**
 - **On Break**
4. Check that your current status updates on the screen.



Example:

- 1 If you are ready to receive assignments, tap Available.
- 2 If you are currently handling an incident, tap Responding.
- 3 If you are temporarily unavailable, tap On Break.

6. Check Your Current Incident

If you have an assigned incident, your dashboard will show a **Current Task**.

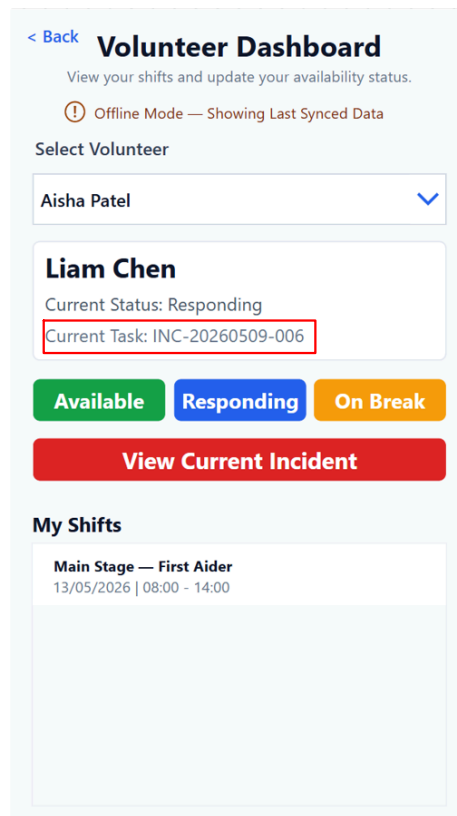
Example:

- 1 Current Task: INC-20260509-006

To view the incident details, tap:

- 1 View Current Incident

Figure 5. View Current Incident Button

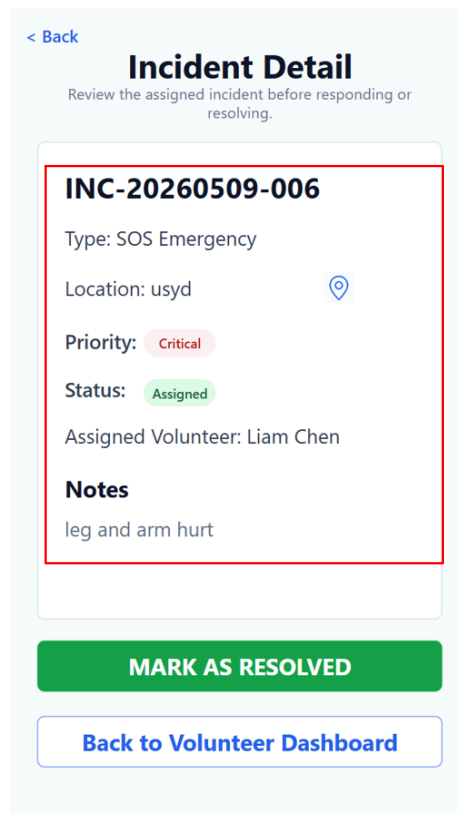


This button opens the **Incident Detail** screen.

7. Review Incident Details

The **Incident Detail** screen shows the emergency information assigned to you.

Figure 6. Incident Detail Screen



You should review the following details before responding:

Field	Meaning
Incident ID	Unique ID of the incident.
Type	Type of emergency, such as SOS Emergency or Heat Stress.
Location	Where the incident happened.
Priority	Urgency level, such as Critical.
Status	Current incident state, such as Assigned.
Assigned Volunteer	Volunteer responsible for this incident.
Notes	Extra details from the resident or administrator.

Example:

```
1 Incident ID: INC-20260509-006
2 Type: SOS Emergency
3 Location: usyd
4 Priority: Critical
5 Status: Assigned
6 Assigned Volunteer: Liam Chen
7 Notes: leg and arm hurt
```

Before moving, make sure the incident is assigned to you and that the location is clear.

8. Mark the Incident as Resolved

After the incident has been handled, return to the **Incident Detail** screen and tap:

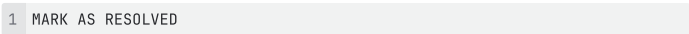
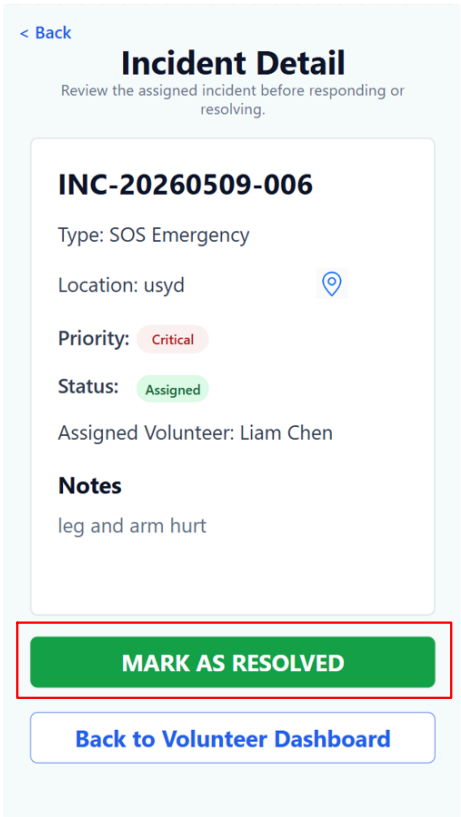


Figure 7. Mark as Resolved Button



When you tap this button, the system updates the incident and your volunteer status.

Expected system update:

Data	Updated Result
Incident Status	Resolved
Volunteer Status	Available
Current Task	Cleared

This tells administrators that the incident has been completed and that you may be available for another task.

9. Return to the Volunteer Dashboard

After marking the incident as resolved, tap:

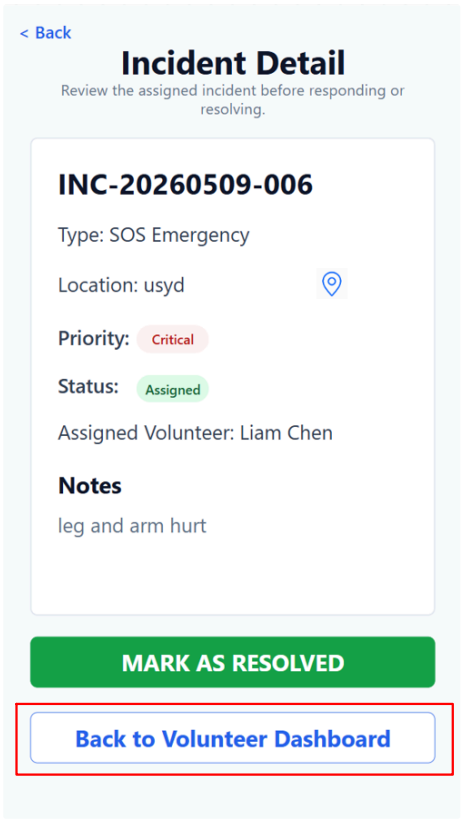
1 Back to Volunteer Dashboard

or use:

1 < Back

You should return to the Volunteer Dashboard and check that your status has updated correctly.

Figure 8. Back to Volunteer Dashboard



10. View Your Assigned Shifts

The **My Shifts** section shows your upcoming duty shifts.

Figure 9. My Shifts Section

[< Back](#)

Volunteer Dashboard

View your shifts and update your availability status.

ⓘ Offline Mode — Showing Last Synced Data

Select Volunteer

Aisha Patel

Liam Chen

Current Status: Responding

Current Task: INC-20260509-006

AvailableRespondingOn Break

View Current Incident

My Shifts

Main Stage — First Aider

13/05/2026 | 08:00 - 14:00

Each shift may show:

```
1 Sector
2 Role
3 Date
4 Start time
5 End time
```

Example:

```
1 Main Stage — First Aider
2 13/05/2026 | 08:00 - 14:00
```

Use this section to confirm where and when you are expected to work.